

Hormones

The electrical signals that zoom along nerves are not the only kinds of messages that travel around the body. Hormones are chemicals that carry messages. They are released into the blood by glands and have powerful effects elsewhere, acting more slowly than nerve signals. Hormones control growth, reproduction, and many other processes.

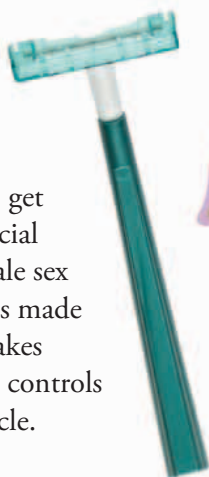
The adrenal glands release the hormone adrenaline, which prepares the whole body for sudden action.

THE ENDOCRINE SYSTEM

The glands and tissues that produce hormones make up the endocrine system. The main endocrine glands are found in the brain, neck, belly, and groin. Many other organs release hormones too, including the heart, stomach, intestines, and even the skin and bone.

SEX HORMONES

The development of male and female characteristics as we grow up is controlled by sex hormones. The male sex hormone testosterone is made by the testes. It causes muscles to get larger and makes facial hair grow. The female sex hormone estrogen is made in the ovaries. It makes breasts develop and controls the reproductive cycle.



In males, the two testes make the sex hormone testosterone.

The pituitary gland makes seven hormones, including growth hormone, which causes bones to grow longer.

The thyroid gland produces thyroxine, which affects the speed of chemical reactions inside cells.

The pancreas makes the hormones insulin and glucagon, which control blood sugar levels.